18.04.24

OPENCAMPUS.sh

Machine Learning With TensorFlow

GENERAL INTRODUCTION

- Personal Introduction
- Intro to opencampus.sh
- Organizational Matters
- Introductory Discussion on AI
- Coursera Registration
- Course Projects
- ML Frameworks

PERSONAL INTRODUCTION

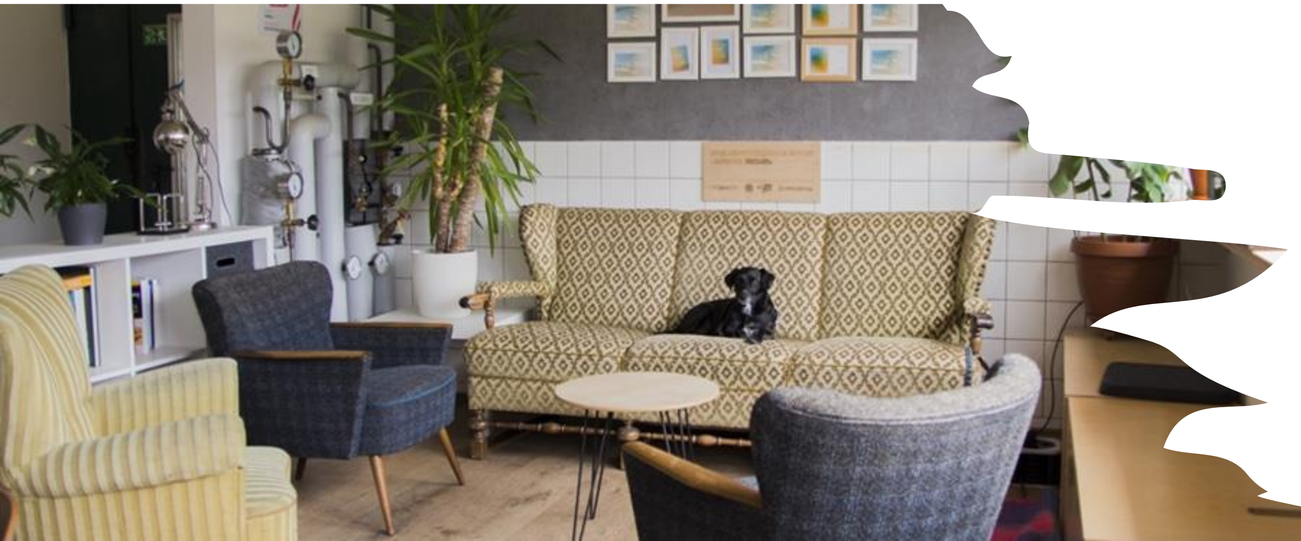
OPENCAMPUS.sh

- Nonprofit organization which oversees a variety of initiatives
- Offering a wide range of educational opportunities, support, and networking for entrepreneurs, creatives, and anyone curious, regardless of age, educational background, or origin
- The services are open to everyone and mostly free.
- The goal is to support the entrepreneurial landscape, promote creative change processes, and contribute to innovative and sustainable future development.









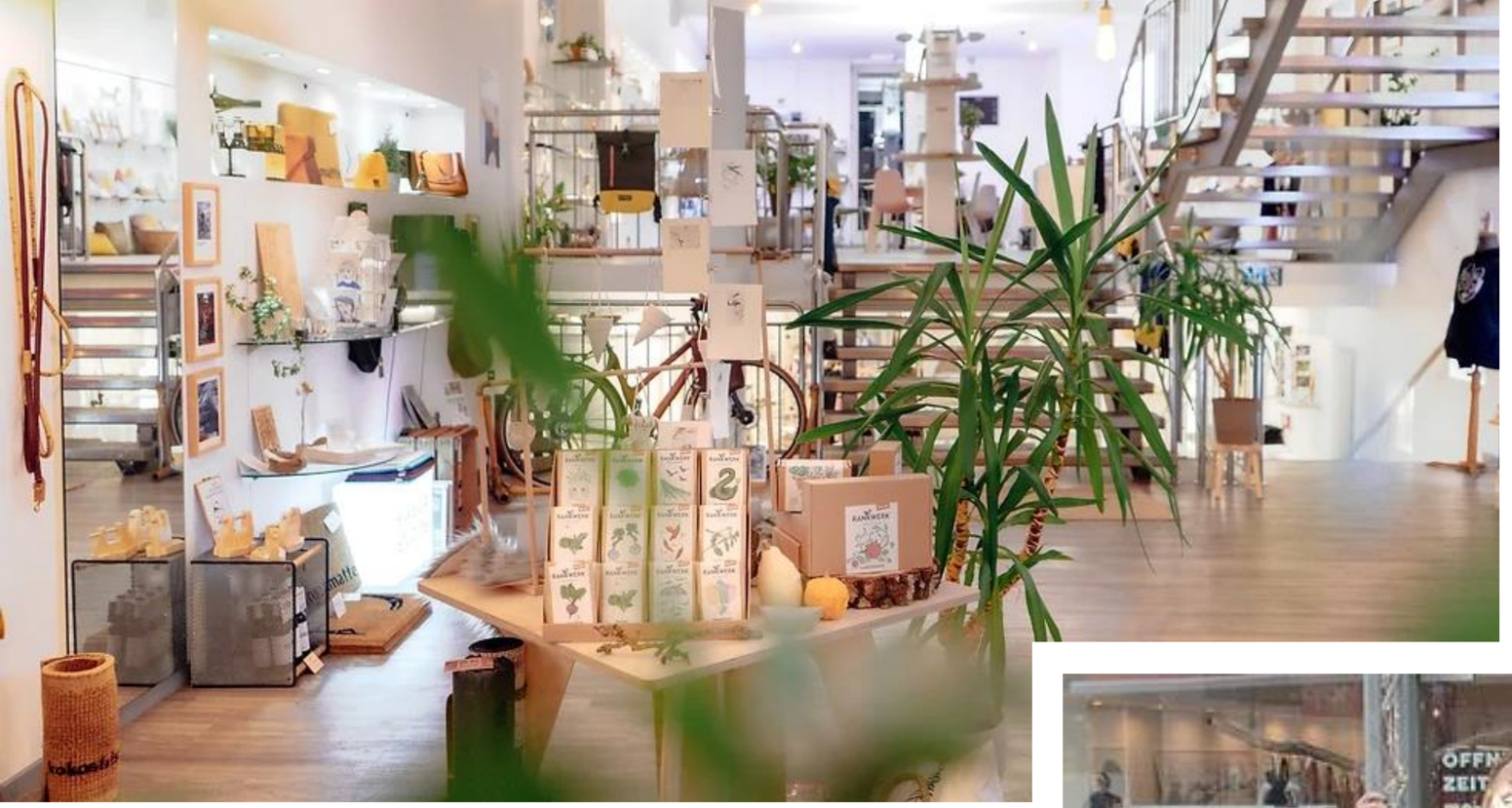
starter kitchen.de



CoBL

COZY WORKING, CULTURE & EVENTS





KOSMOS
by opencampus.sh



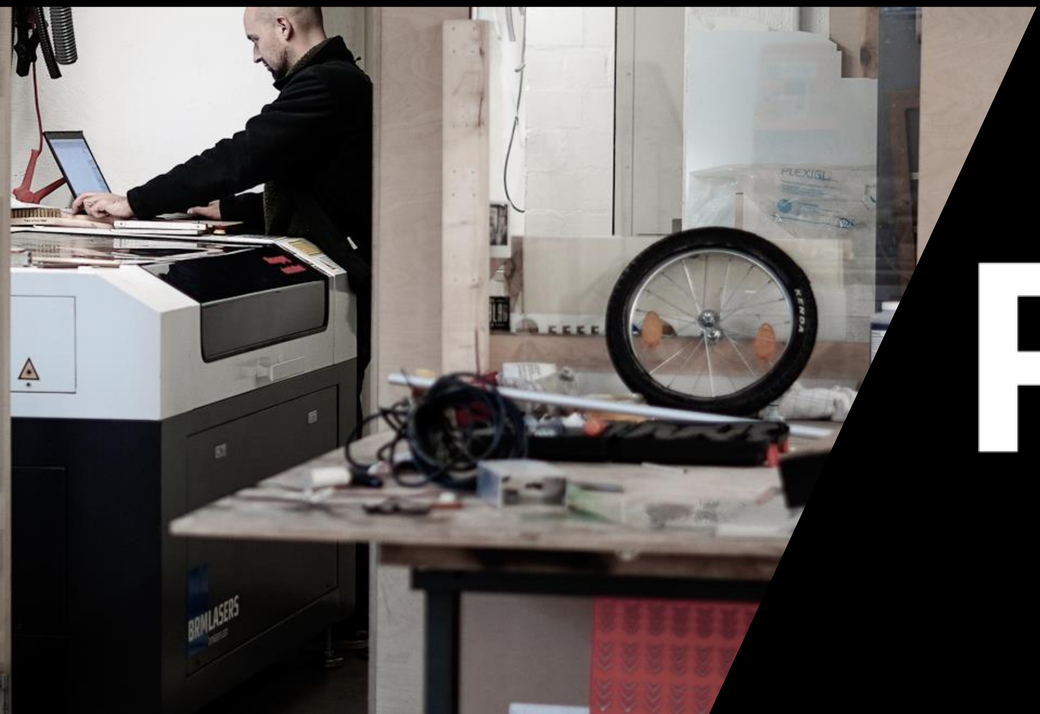












FABLAB KIEL





MFG 5 KIEL WE CONNECT. 13.–14. JUNI 2024

WATERKANT FESTIVAL

WATERKANT.SH









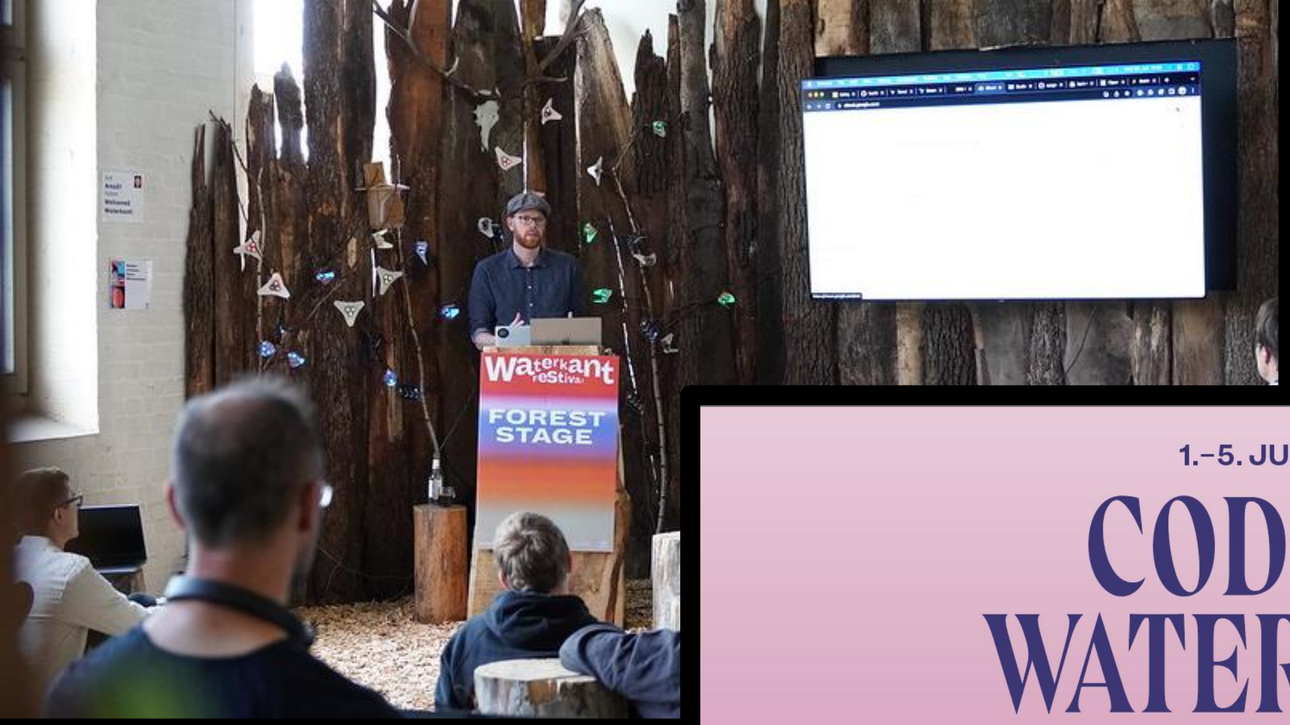






WATERKANT EXHIBITION 2024

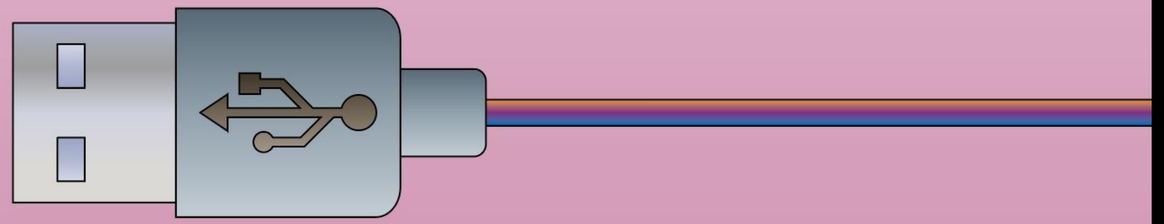




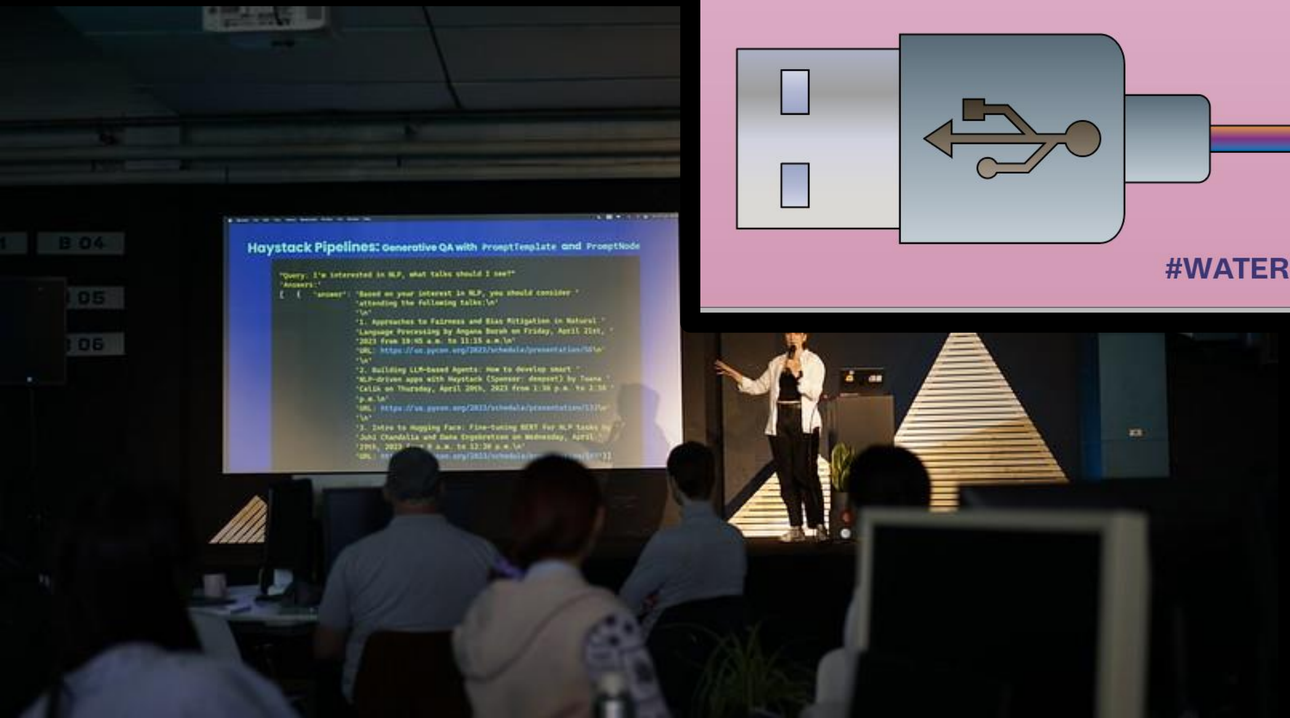


1.–5. JULI 2024

CODING. WATERKANT



#WATERKANT24





1.–5. JULI 2024

CODING. WATERKANT

- Work for four consecutive days:
  - on your own machine learning project or
  - support others in their project
- Take part in workshops

- Get input and feedback by invited experts
- Present your work to a larger audience.
- Take educational leave
- Get accommodation on site

https://coding.waterkant.sh

https://kiel.ai

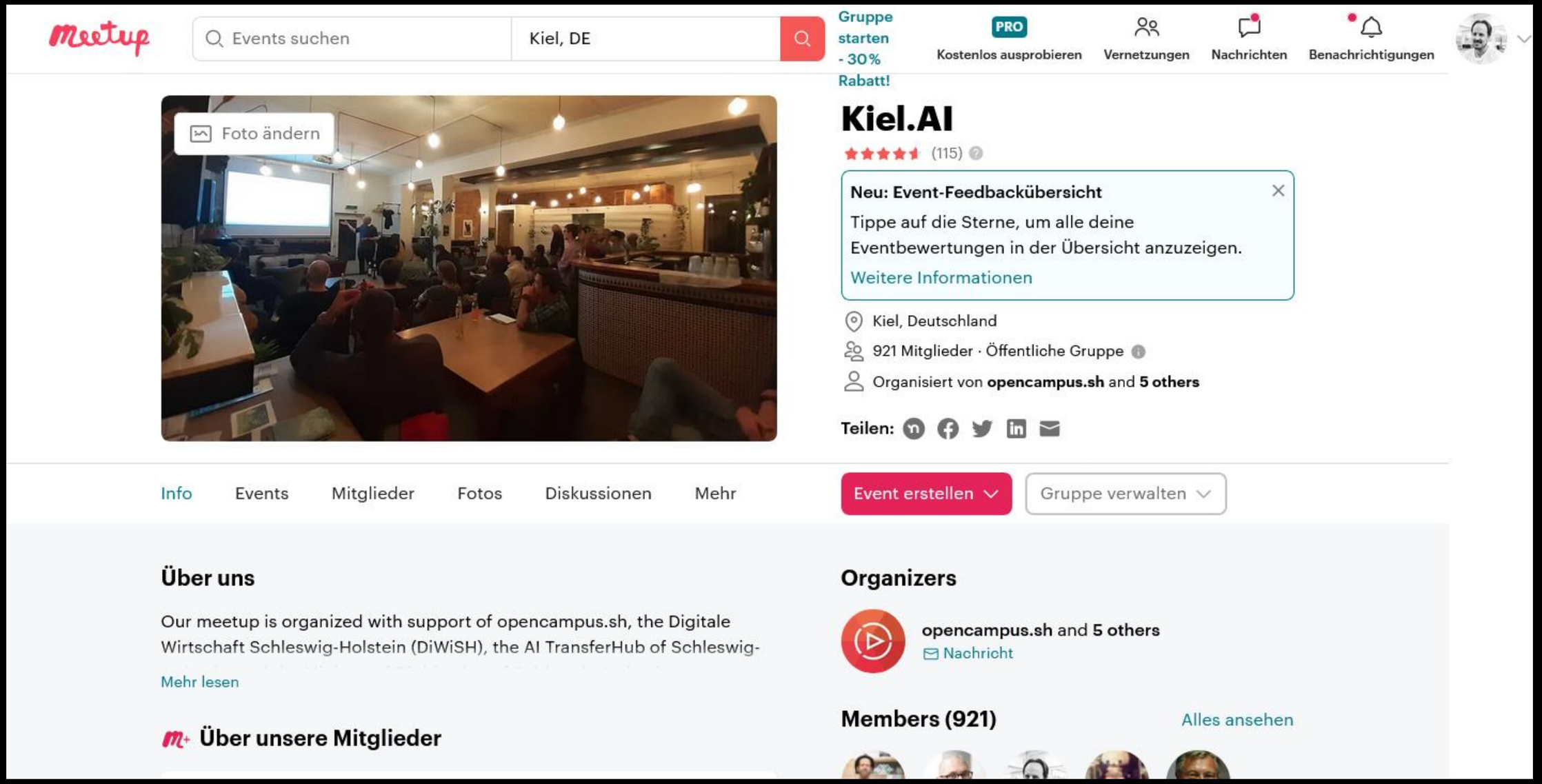
ML Degree Meetup Coding.Waterkant Chat



Kiel.AI

OPENCAMPUS.sh

DiWiSH
DIGITALE WIRTSCHAFT
SCHLESWIG-HOLSTEIN
CLUSTERMANAGEMENT



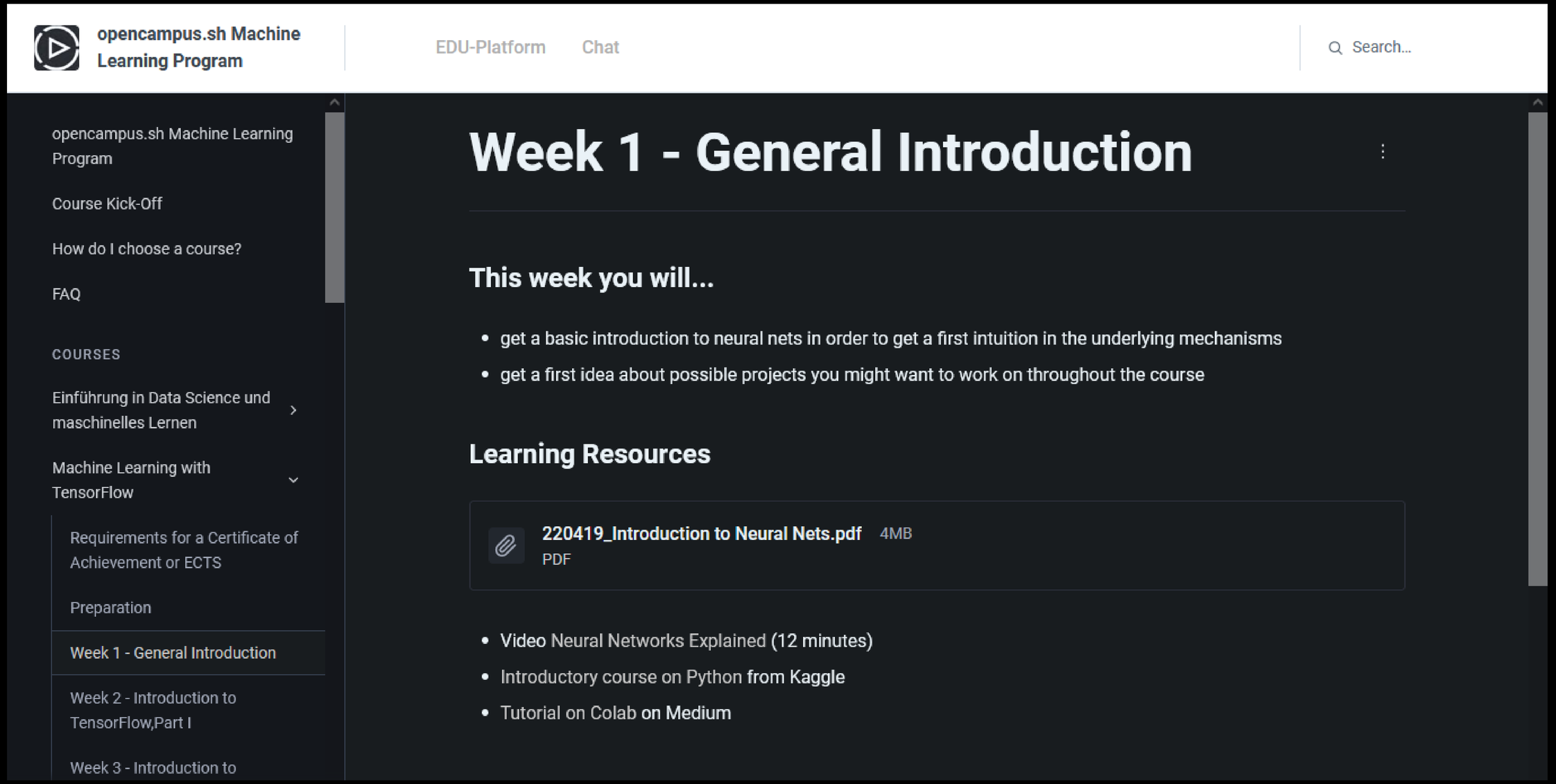
https://www.meetup.com/kiel-ai/

CHAT



- **Please, ask questions to us in the chat**

COURSE HANDBOOK



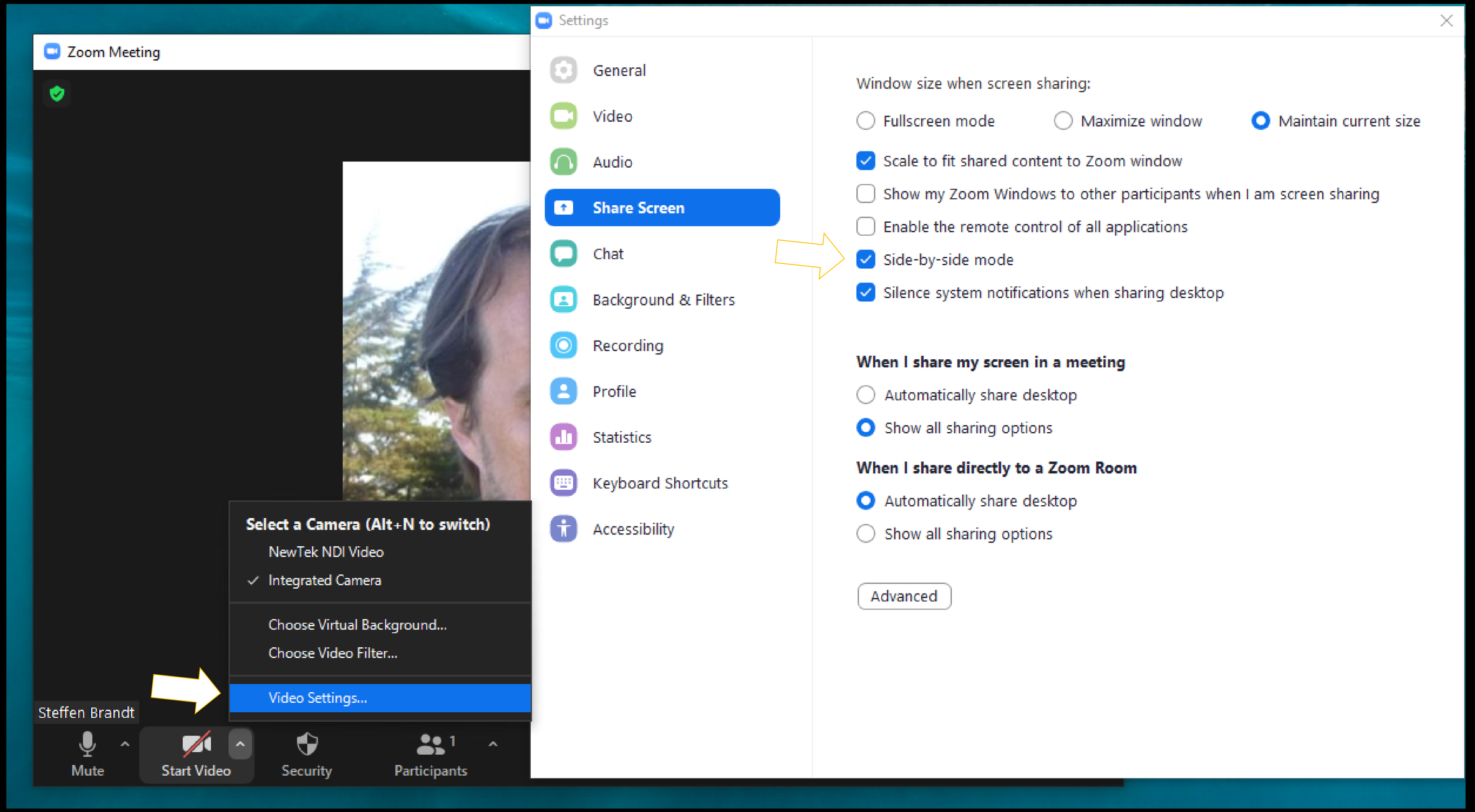
ORGANIZATIONAL MATTERS

- Use your full names in the zoom meetings!
- Scan the QR-Code if you participate in presence
- Complete your profile in the Mattermost chat with your full name and a photo.
- Please write us if you will not go on with the course!

ZOOM

- Try the different viewing modes:
  - Gallery View/ Active Speaker
  - Split Screen/ Full Screen Mode

- Maybe watch this video to get an idea: https://www.youtube.com/watch?v=v3IPAbpVjd4


Zoom Meeting
Select a Camera (Alt+N to switch)
NewTek NDI Video
Integrated Camera
Choose Virtual Background...
Choose Video Filter...
Video Settings...
Steffen Brandt
Mute
Start Video
Security
Participants
1
Settings
General
Video
Audio
Share Screen
Chat
Background & Filters
Recording
Profile
Statistics
Keyboard Shortcuts
Accessibility
Window size when screen sharing:
Fullscreen mode
Maximize window
Maintain current size
Scale to fit shared content to Zoom window
Show my Zoom Windows to other participants when I am screen sharing
Enable the remote control of all applications
Side-by-side mode
Silence system notifications when sharing desktop
When I share my screen in a meeting
Automatically share desktop
Show all sharing options
When I share directly to a Zoom Room
Automatically share desktop
Show all sharing options
Advanced

18.04.2024 Introduction
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

25.04.2024 Introduction to TensorFlow for AI, Machine Learning, and Deep Learning, Part I
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

02.05.2024 Introduction to TensorFlow for AI, Machine Learning, and Deep Learning, Part II
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

09.05.2024 Project Work
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

16.05.2024 Natural Language Processing, Part I
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

23.05.2024 Natural Language Processing, Part II
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

30.05.2024 Convolutional Neural Networks, Part I
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

06.06.2024 Convolutional Neural Networks, Part II
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

13.06.2024 Sequences, Time Series and Prediction, Part I
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

20.06.2024 Sequences, Time Series and Prediction, Part II
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

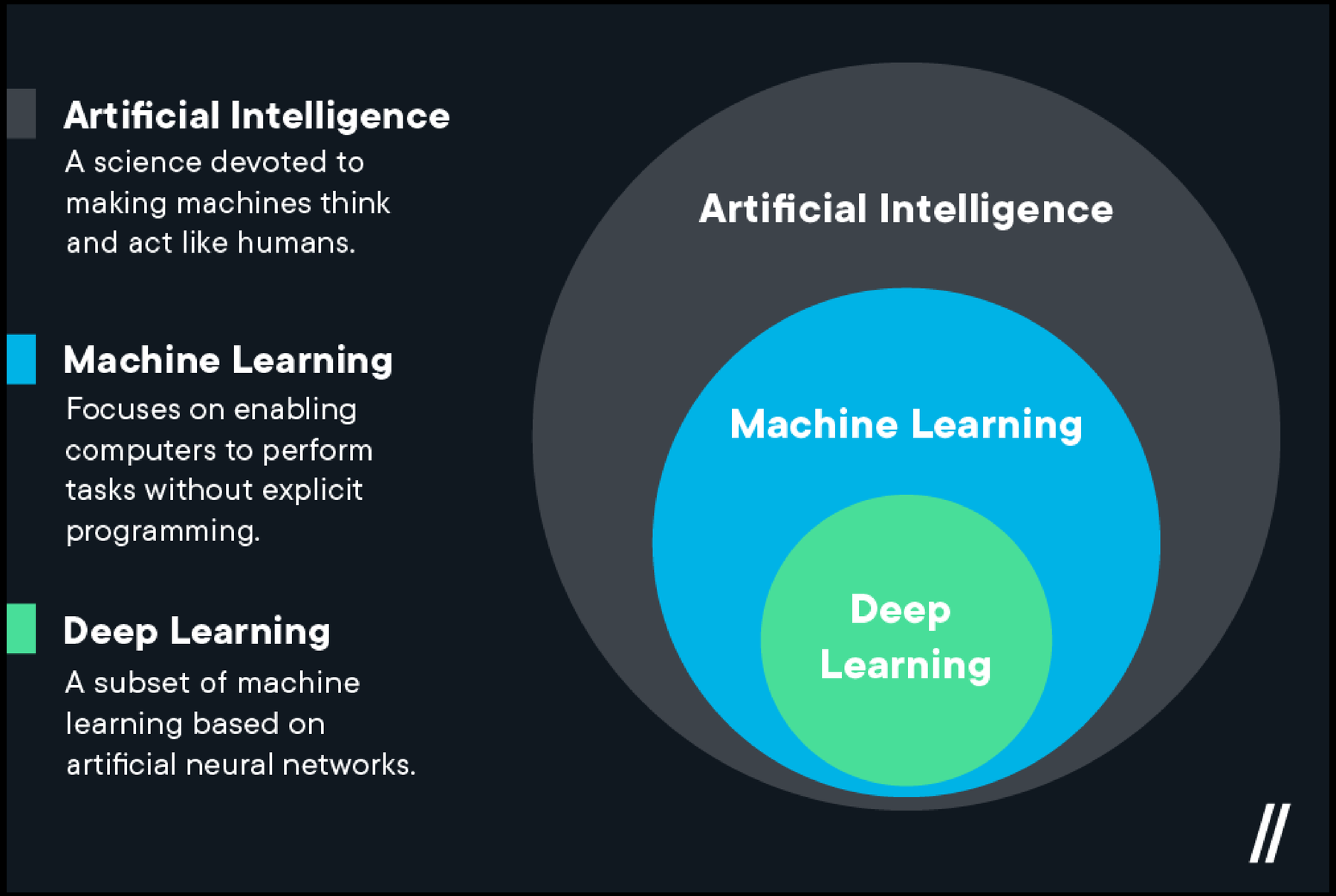
27.06.2024 Presentation of the Final Projects, Part I
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

04.07.2024 Presentation of the Final Projects, Part II
18:15 - 20:00 Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

Coding.Waterkant

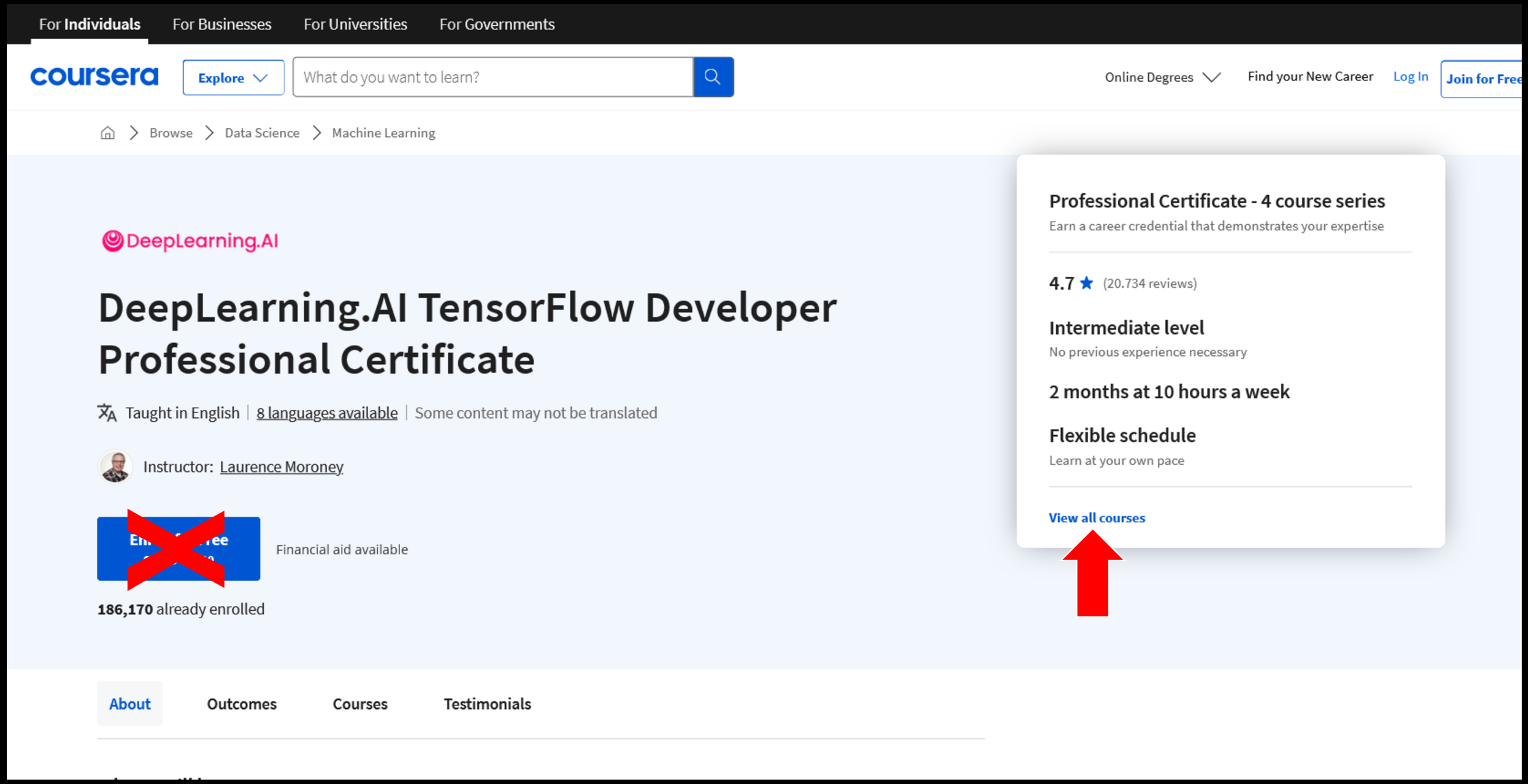
FIRST BREAKOUT

- 15-20 Minutes
- Present yourself
- Discussion Questions:
  - Do you know examples for Machine Learning?
  - Do you know examples for Deep Learning?



Deep Learning vs. Machine Learning—What's the Difference? (2021, February 8). Flatiron School. https://flatironschool.com/blog/deep-learning-vs-machine-learning/

DO NOT SIGN UP FOR THE SPECIALIZATION



CLICK ON THE INDIVIDUAL COURSE

Professional Certificate - 4 course series

TensorFlow is one of the most in-demand and popular open-source deep learning frameworks available today. The DeepLearning.AI TensorFlow Developer Professional Certificate program teaches you applied machine learning skills with TensorFlow so you can build and train powerful models.

Read more



Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

Course details

Course 1 • 17 hours • **4.8** ★ (18.883 ratings)

What you'll learn

- ✓ Learn best practices for using TensorFlow, a popular open-source machine learning framework
- ✓ Build a basic neural network in TensorFlow
- ✓ Train a neural network for a computer vision application
- ✓ Understand how to use convolutions to improve your neural network

Skills you'll gain

Computer Vision | Tensorflow | Machine Learning

Convolutional Neural Networks in TensorFlow

Course 2 • 16 hours • **4.7** ★ (7.912 ratings)

Instructor

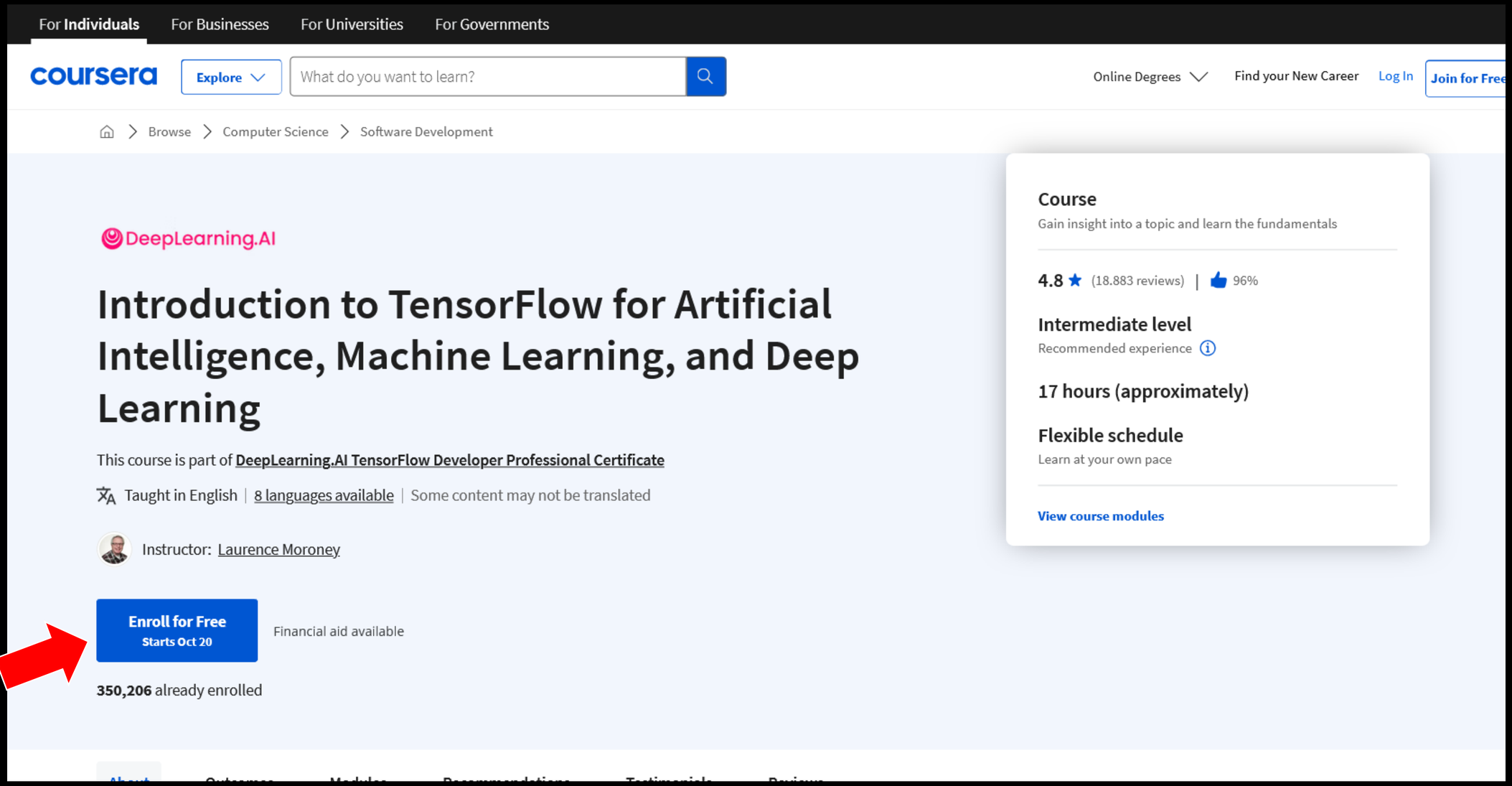


Laurence Moroney
DeepLearning.AI
15 Courses • 456,481 learners

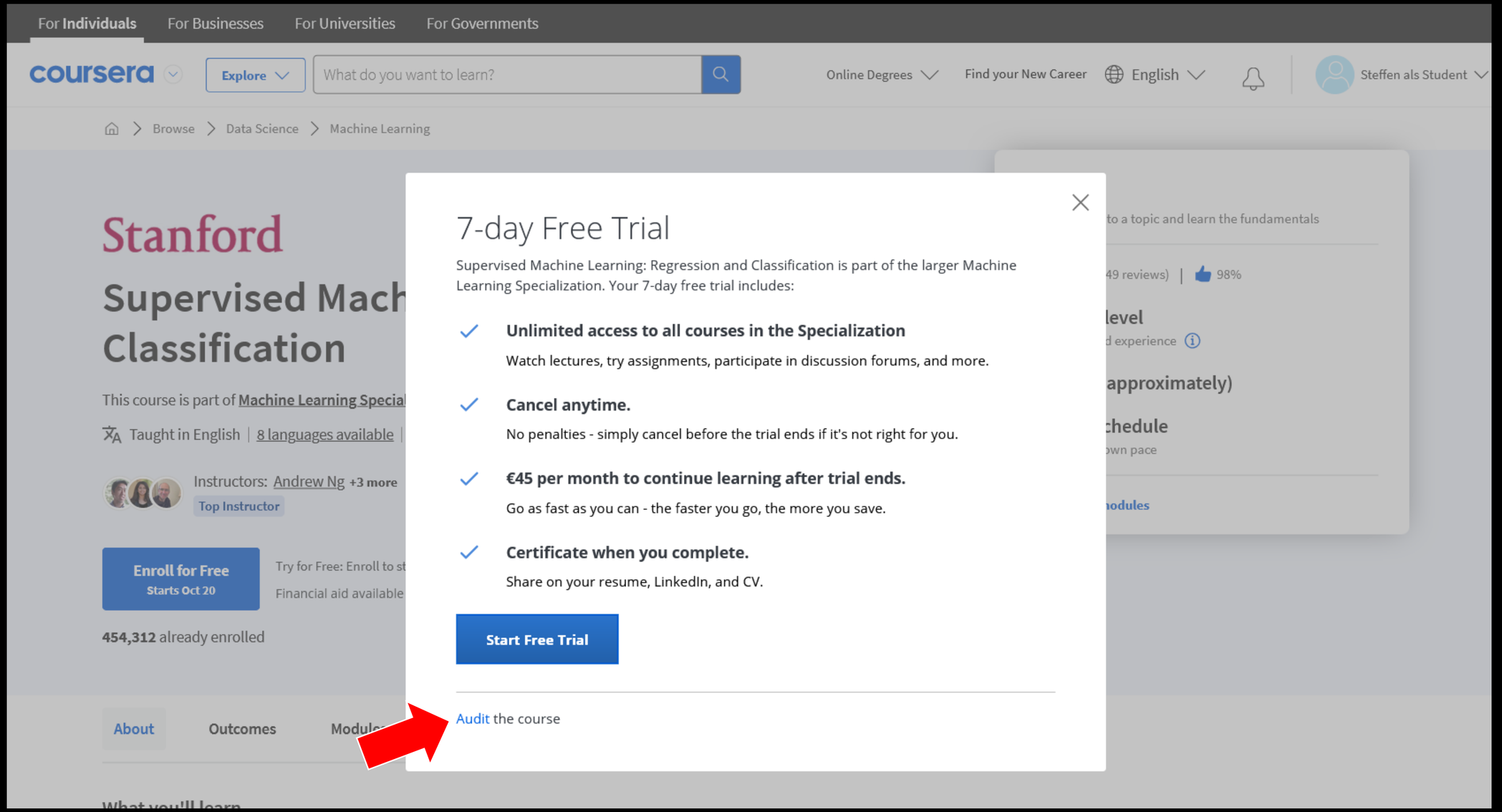
Offered by

DeepLearning.AI
Learn more

ENROLL FOR INDIVIDUAL COURSE



SELECT AUDIT MODUS



EXERCISES

- Each week two to four of you will present the exercises given in the course handbook
- Each of you presents at least once

EXERCISES:
WHO WILL PRESENT NEXT WEEK?

PROJECTS

Option 1:

Bring your own idea and data

Option 2:

Do one of the two default projects

RELEVANCE OF THE PROJECTS

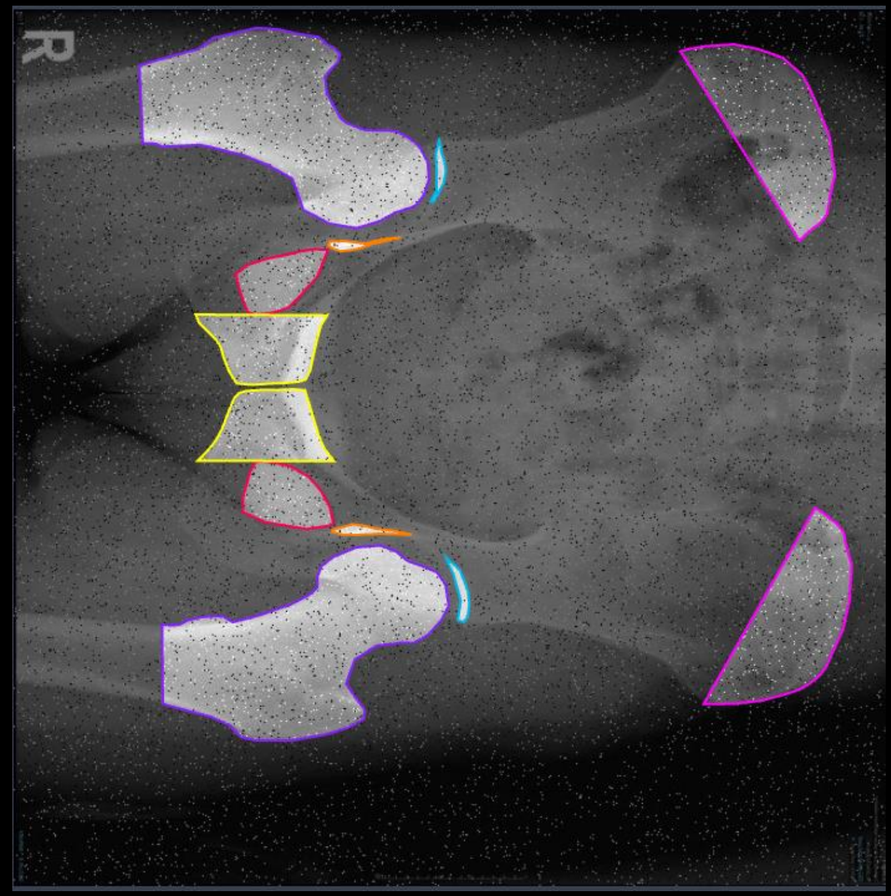
- Most important for a career in ML will be work experience and your GitHub profile
- Focus on building a noteworthy GitHub project repository
- Use the template repository
- Outstanding projects will be nominated for the VDE prize

SEGMENTATION OF THE THIGH BONE (FEMUR) ON PELVIC X-RAY IMAGES

- **Partner:**
  i²Lab at the UKSH

- **Goal:**
  The goal is to use artificial neural networks, like U-Net, to automatically segment the femur and five other anatomies on clinical hip X-rays, specifically pelvis AP X-rays.

- **Data:**
  - 2D X-ray images (jpeg)
  - 458 images of 3000 by 3000 pixels.
  - Segmentation masks for six different classes (anatomies).

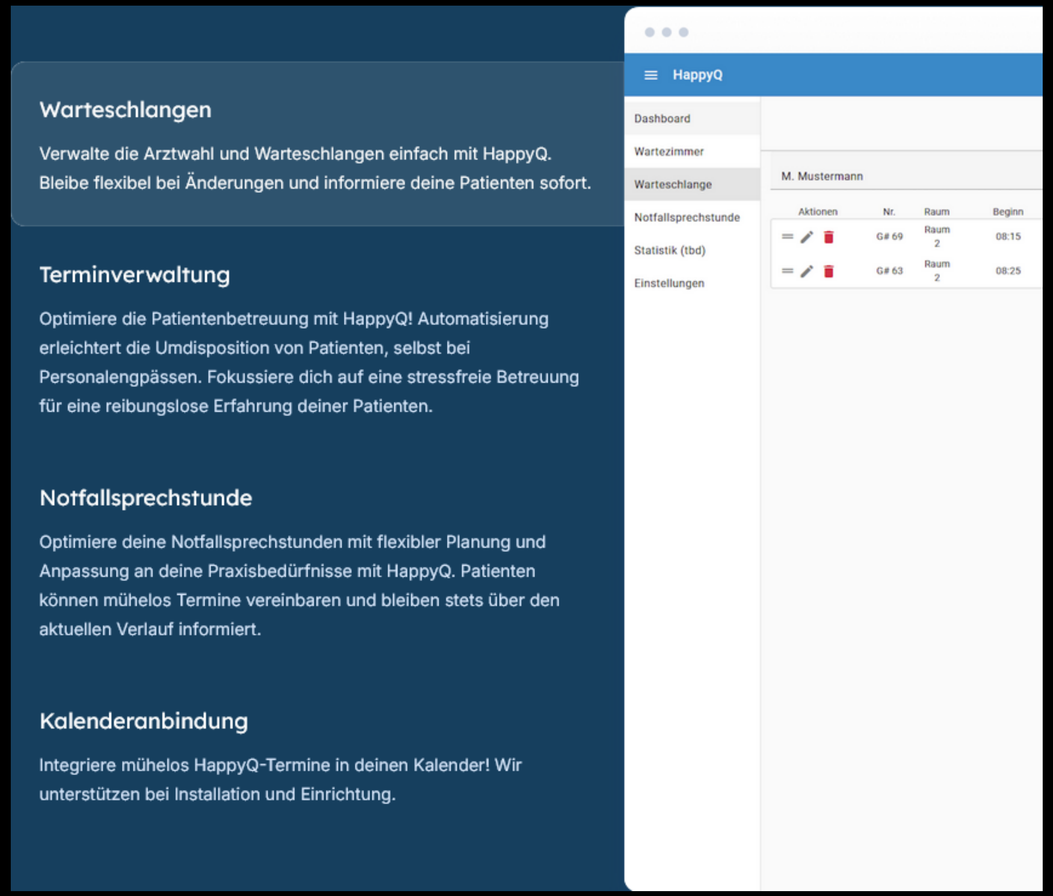


PREDICTION OF PATIENT TREATMENT TIMES

- **Partner:**
  MintCode Solutions with their product HappyQ

- **Goal:**
  …

- **Data:**
  …

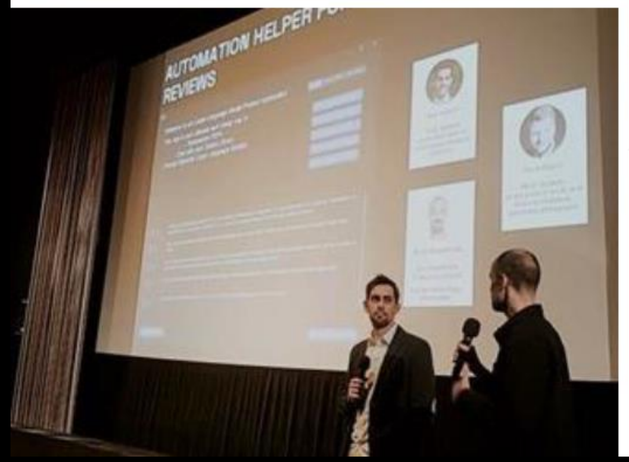


VDE

SPECIAL PRIZE MACHINE LEARNING













COURSES

Fine-Tuning and Deployment of Large Language Models

Archive

EVENTS

Coding.Waterkant 2023

Prototyping Week

PROJECTS

How to Start, Complete, and Submit Your Project

Possible Projects

Past Projects

ADDITIONAL RESOURSES

Glossary

Coursera

Selecting the Optimizer

Choosing the Learning Rate

Learning Linear Algebra

Learning Python

Support Vector Machines

ML Statistics

TOOLS

Git

RStudio

Google Colab

Zoom

Powered by GitBook

How to Start, Complete, and Submit Your Project

Starting Your Project

Working on Your Project

Submitting Your Project

Was this helpful?

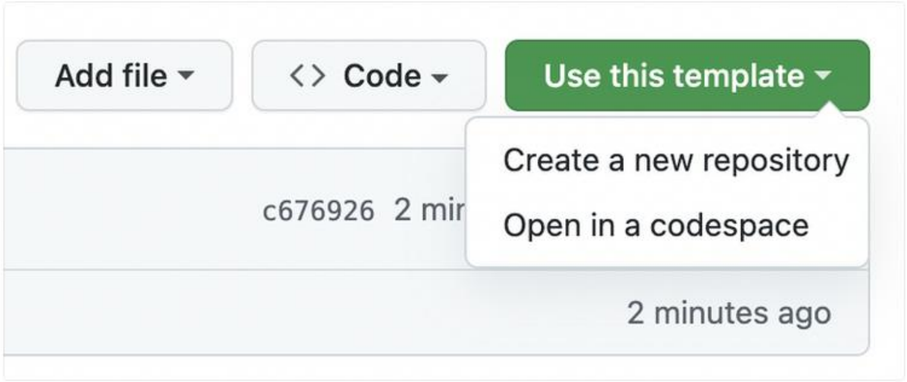
Export as PDF

In all Machine Learning courses you have:

- to complete a machine learning project in a team of up to 4 participants,
- attend at least all but 2 sessions of the course, and
- use the provided project template repository for documentation (unless otherwise instructed).

Starting Your Project

1. **Navigate to the** Template Repository
2. **Use this Template**: Above the file list, click the "Use this template" button.



Use this template button

3. **Create Repository from Template**: You'll be prompted to name your new repository and you can choose whether it should be public or private. You'll also have the option to include all branches in the template repository, if there are more than one.
4. **Create Repository**: Click "Create repository from template" to create the new repository.
5. **Clone the New Repository**: You can now clone the new repository to your local machine using `git clone` and start working on your project.

Working on Your Project

PROJECT INTERESTS

GOOGLE COLAB

Coding.Waterkant Hackathon
Prototyping Week

PROJECTS
Requirements
Possible Projects
Past Projects

ADDITIONAL RESOURSES
Glossary
Coursera
Selecting the Optimizer
Choosing the Learning Rate
Learning Linear Algebra
Learning Python

TOOLS
Git
RStudio
Google Colab
Zoom

Google Colab

Programming Environment to develop and train Neural Networks in the Google Cloud

Colab is an online **programming environment** for Python including a free runtime with a GPU and is the standard programming environment for our machine learning projects.

The following blog article provides you with a great written introduction to Google Colab:

Google Colab 101 Tutorial with Python—Tips, Tricks, and FAQ

An in-depth tutorial on how to use Google Colab with Python, along with Colab's tips, tricks, and FAQ

medium.com



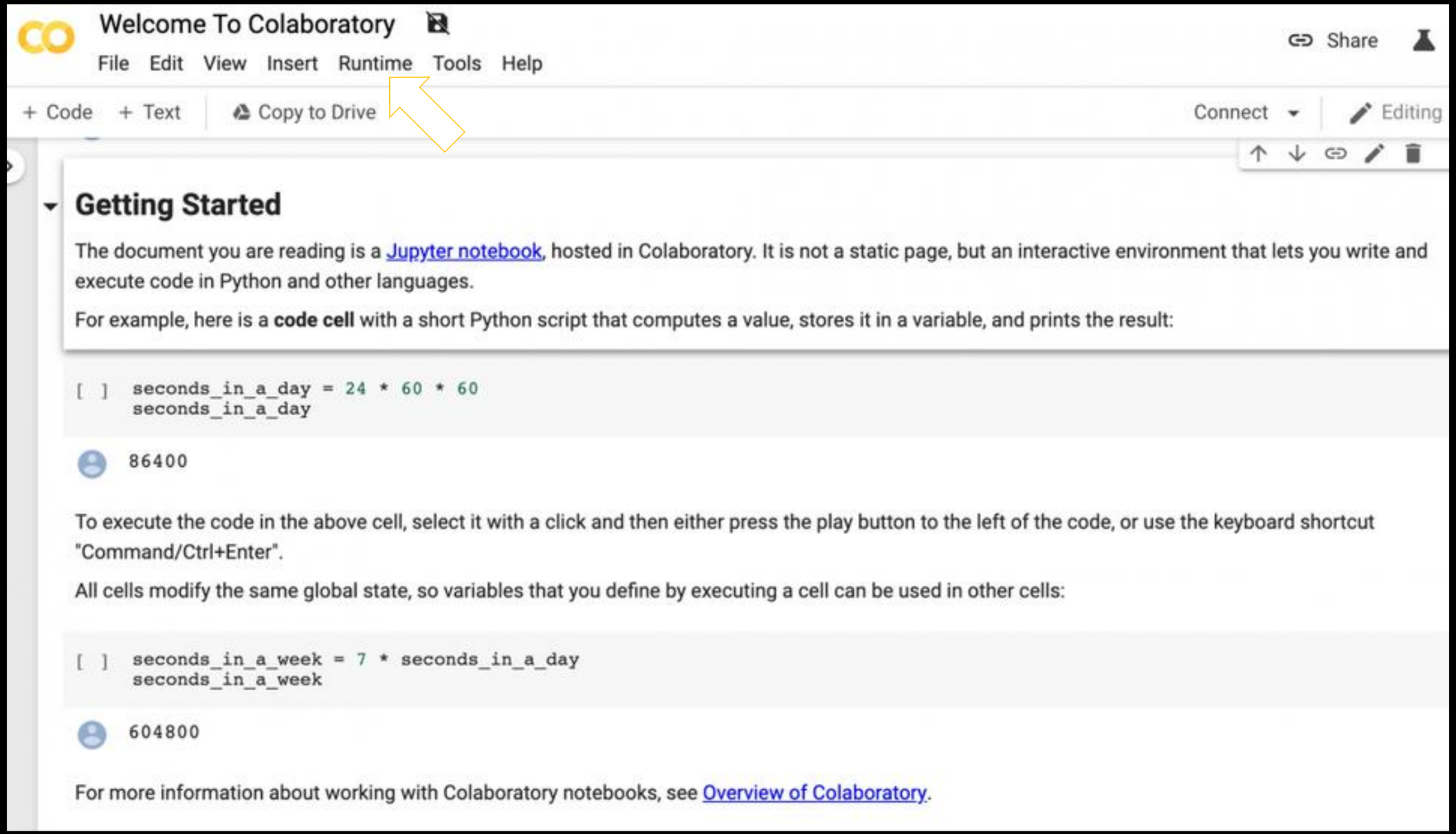
If you want to get a little bit deeper into using Colab, make sure to also read this blog article:

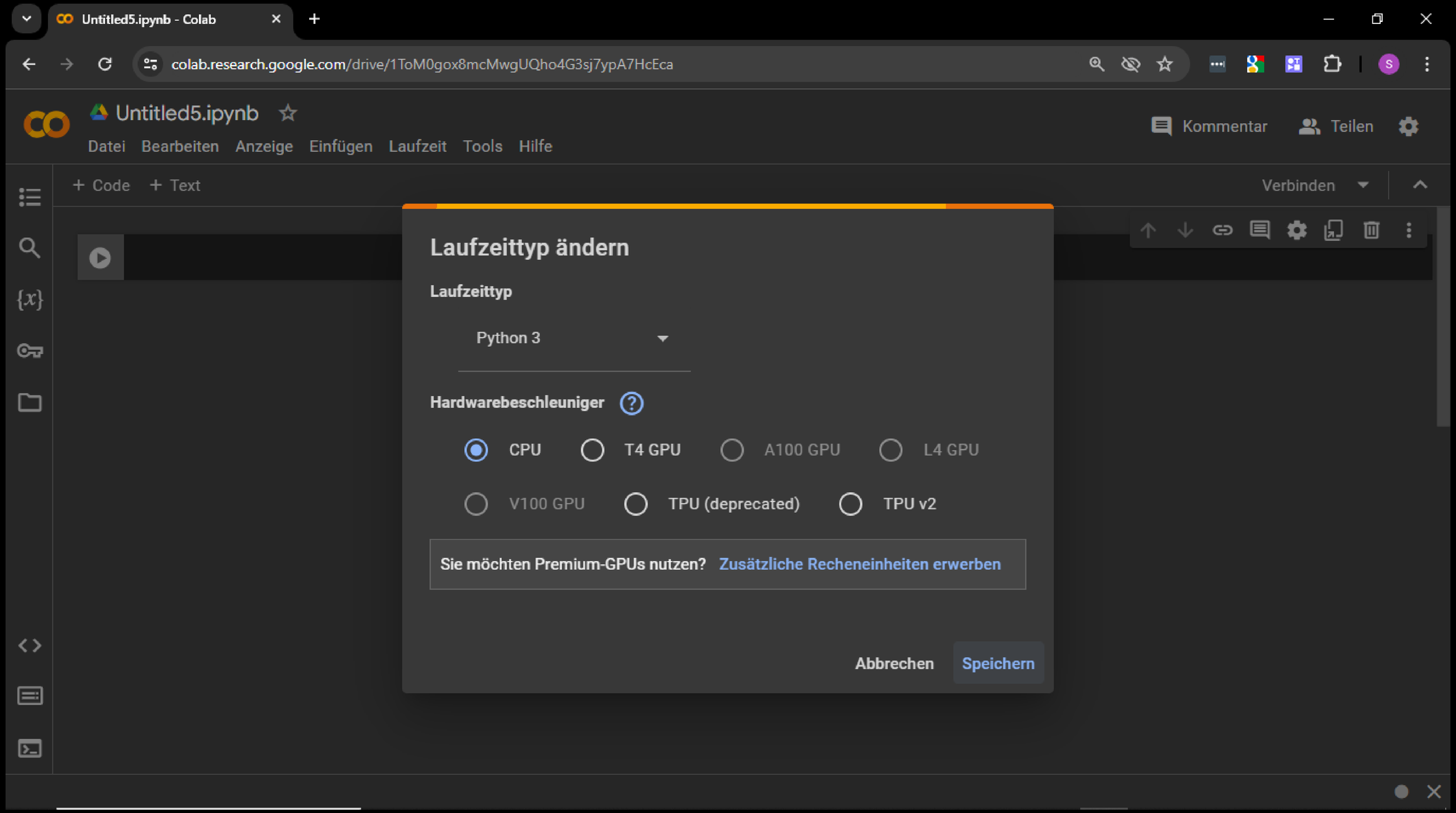
Configuring Google Colab Like A Pro

How to Do Research Quality Machine Learning on a Budget

medium.com




Welcome To Colaboratory
File Edit View Insert Runtime Tools Help
Share
+ Code + Text Copy to Drive
Connect
Editing
Getting Started
The document you are reading is a Jupyter notebook, hosted in Colaboratory. It is not a static page, but an interactive environment that lets you write and execute code in Python and other languages.
For example, here is a code cell with a short Python script that computes a value, stores it in a variable, and prints the result:
[ ] seconds_in_a_day = 24 * 60 * 60
seconds_in_a_day
86400
To execute the code in the above cell, select it with a click and then either press the play button to the left of the code, or use the keyboard shortcut "Command/Ctrl+Enter".
All cells modify the same global state, so variables that you define by executing a cell can be used in other cells:
[ ] seconds_in_a_week = 7 * seconds_in_a_day
seconds_in_a_week
604800
For more information about working with Colaboratory notebooks, see Overview of Colaboratory.


Untitled5.ipynb - Colab
colab.research.google.com/drive/1ToM0gox8mcMwgUQho4G3sj7ypA7HcEca
Untitled5.ipynb
Datei Bearbeiten Anzeige Einfügen Laufzeit Tools Hilfe
Kommentar
Teilen
+ Code + Text
Verbinden
Laufzeittyp ändern
Laufzeittyp
Python 3
Hardwarebeschleuniger
CPU
T4 GPU
A100 GPU
L4 GPU
V100 GPU
TPU (deprecated)
TPU v2
Sie möchten Premium-GPUs nutzen? Zusätzliche Recheneinheiten erwerben
Abbrechen
Speichern

ML LIBRARIES (LOWER LEVEL)







PYTORCH

ML LIBRARIES (HIGHER LEVEL)



Transformers

spaCy



haystack
by deepset



LangChain

LUDWIG



Auto gpt

DEVELOPMENT ENVIRONMENTS





Visual Studio Code









TASKS UNTIL NEXT WEEK

- Completion of the learning material of week 1 and 2 of the course "introduction to TensorFlow“
- Complete the two assignments given in the following notebooks:
  - Assignment Notebook 1
  - Assignment Notebook 2

  (Who presents?)
- Bring questions!